

Thaddeus Stevens College of Technology

CPU Architecture

Joseph Costello

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Mr. McLaughlin

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CPU architecture is the way a CPU's overall design is described. Within architecture is the contrast between hardware and software, specifications about the instruction set architecture (ISA), and microarchitecture, which describes the internal organization of the CPU, including the interior components such as the control unit, registers, cache hierarchy, memory management unit (MMU), and arithmetic logic unit (ALU). Both the architecture, and further microarchitecture, are what controls CPU's performance, power efficiency, and how it operates as a whole. Some CPU architectures that are common right now are x86, used by both intel and AMD, ARM, which is used in Apple's M1/M2 chips, and RISC-V, which is open-source and has been gaining popularity due to it being free to use and customize.

CPUs work by receiving and executing instructions from the computer's memory. The instructions are telling the CPU what it needs to do with a specific item stored on the computer. When the instruction is sent from the memory, it first goes through the control unit, which decodes the instruction as well as deciding on any needed data items to execute the instruction properly. From the control unit, the data then moves on to the ALU, which is what actually carries out the instruction, and stores values back into the memory. CPUs today can handle many instructions quickly and easily due to how far technology has come. However, in the early days of computers, instructions were not always executed as smoothly or easily.

CPUs started in the 1950's when every computer was fully unique due to a lack of mass production, meaning each part in a computer or CPU was made just for that specific device. One of the few things stopping freedom of people designing these things, was costs at the time being very expensive for electronics. This led to only basic features being added into these early and unique computers. Near the end of the 1950's there were finally factory constructed computers which led to more computers being installed helping with innovation. In the 1960's a major

development happened, IBM started to make computers that would be able to run the same programs on different computers, which would have different performances for different prices. They would all be based off one reference computer named the System/360 (S/360). This meant that they made multiple different CPUs for the different form factors. Cheaper computers would have a much simpler processor, compared to the more expensive models' processors. The computers would then use a microcode emulator to be able to run the programs. The bigger processors had smaller emulators leading to faster performance. The S/360 was impressively complex and capable for the time and was designed to support multiple forms of processing as it was intended to be a follow up to multiple types of machines.

The 1970's also had incredible amounts of innovation similar to the decades before it. 1971 was the year of the first commercially released microprocessor, the Intel 4004. It was binary coded and would soon be followed by the Intel 8008, an 8-bit microprocessor compared to the 4004's 4 bits. The 8008 was then followed by, likely the main reason that microprocessors were so popular, the Intel 8080. The 8080 originally followed in the 8008's footsteps being used in small devices that didn't require large amounts of processing, however it quickly became a target CPU by being the standard operating system for personal computers at the time. This innovation as well as the lowering prices due to transistors becoming more affordable, led to 'minicomputers' becoming popular. The most powerful minicomputer of the time was the 16-bit PDP-11 minicomputer, until 32-bit minicomputers started to be produced. The Intel 8080 still has relevance today, as it was the blueprint for the 16-bit Intel 8086, which led into the modern day x86 chip type, and every instruction of the 8080 has an equivalent in the x86 instruction set.

The early 1980's had a significant innovation in CPUs as well, reduced instruction set computing (RISC) designs. RISCs were made due to researchers at IBM and UC Berkely

noticing a trend in computers at the time. They noticed that most computer language interpreters and compilers would not use most of the available instructions on machines at the time, leading to a large part of the CPU's power not being properly used. To innovate from this discovery, they started to make RISC computers. RISCs would have a smaller number of instructions while having a larger number of registers, which would overall make a simpler CPU core that could run at a higher speed while only dropping the capabilities that were rarely used anyway. RISCs had multiple approaches that companies would use to try and most effectively run the computers. One of the most common approaches was the 'Harvard architecture.' The Harvard architecture designed machines, the program being used and the data would use different memory devices at the same time which would avoid a problem that would happen in some other programs which would have data and programs in the same memory device, sometimes leading to a 'bottleneck.' One problem with RISCs that led to them being slower than intended was the programs running on them tending to be longer due to compilers having to make longer instruction sets using the simple instructions instead of being able to condense it by using the instructions other computers had at the time. In the early 1990's, it was found by a Japanese company Hitachi that instruction sets could be compressed down to fit in smaller memory systems. This led to RISCs finally being able to be made in a slightly smaller form factor than before, and compressed RISCs are used commonly in applications that don't need to run any older binary software. RISCs are still important in today's CPUs as some new x86 processors internally change instructions into a set that is similar to RISC.

Another progression that was happening in CPUs at the same time as RISCs was companies starting to exploit and work around instruction-level parallelism (ILP). One of the first ways they did this was in the mid 1980's, when they began using instruction pipelining. In

instruction pipelining, the processor will work on different instructions at the same time while they are in different stages of their process. This continues to be a thing in modern CPUs which can do so much more efficiently. IBM actually used pipelining in the late 1950's, but it took until the 1980's for this to become a practice in commercially available CPUs. A similar idea also started to become popular around the same time, which was to process more than one instruction at a time on separate ALUs. For this to work the CPU would find similar instructions that were not dependent on each other and execute them at the same time. These were both limited by ILP due to some programs having a low number of instructions that were not dependent on each other, making it so that neither was always the most effective. One of the bigger innovations of the 1990's was out-of-order execution in which the CPU will execute instructions in an order that's based on the availability of the components. This way the CPU isn't wasting its power by being idle. Another similar concept that can help the CPU to avoid being idle is speculative execution, which is when a CPU will execute instructions from a branch direction before it knows the intended instructions. If the predicted instructions end up being the same as the intended, then the CPU saves processing power and time. This means that an accurate branch predictor is the most important component to make speculative execution work. All these innovations are parts of what has led to CPUs today being so efficient in executing commands quickly.

All of those innovations happened in or before the early 1990's, however some of the most talked about innovations have been more recent than that. One of the more recent and significant CPU innovations is multi-threading. Multi-threading is what allows CPUs to effectively run multiple programs at the same time. Without multi-threading CPUs would need to use context switching which is significantly less efficient. Another large innovation that makes it

possible for computers to run more programs at the same time is multi-core CPUs. Multi-core CPUs split tasks up across multiple cores on one CPU to allow tasks to be run more efficiently without increasing power consumption and heat.

These innovations have all led CPUs to having the capabilities they have today. However, CPUs will continue to evolve as time goes on. One interesting feature in some more recent CPUs is open source processors. Open source processors are a more recent thing, having started to be produced within the last decade. These processors allow people to see and edit the software within CPUs. I think this could be a next step for CPU innovation as people want to have as much control and customization over their devices as possible. I could therefore see people who build their own PCs being interested in this to have more control over every component of their PC. I don't see these ever becoming the standard as a likely thing, due to a lot of people just wanting to not have to think about their CPU that much, as well as potential costs being more on these, however I do think it's a very interesting progression of CPUs and I could definitely see it turning into an interesting sub category of the industry.

Overall, it's hard to say where CPU progression will continue to grow from here as there are already CPUs that are unimaginably strong compared to the first CPUs made. Innovations will continue to happen, however I don't think there will be as fast of progression as there was when they first became a thing as we've almost reached a plateau of how powerful they can realistically, and cost effectively become.

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